# EXPERIMENT 1.1

## Mapped Course Outcomes- CO1

**CO1:** Understand the working of routers, switches and hubs

**AIM:** Understand the working of following (i) IP Address. (ii) Cisco IOS. (iii) Straight Cable & Cross Cable, RJ45 (iv) Layer 2 Switch. (v) Router.

## Apparatus required:

Cisco Packet Tracer

Objective: - Students will understand concepts of

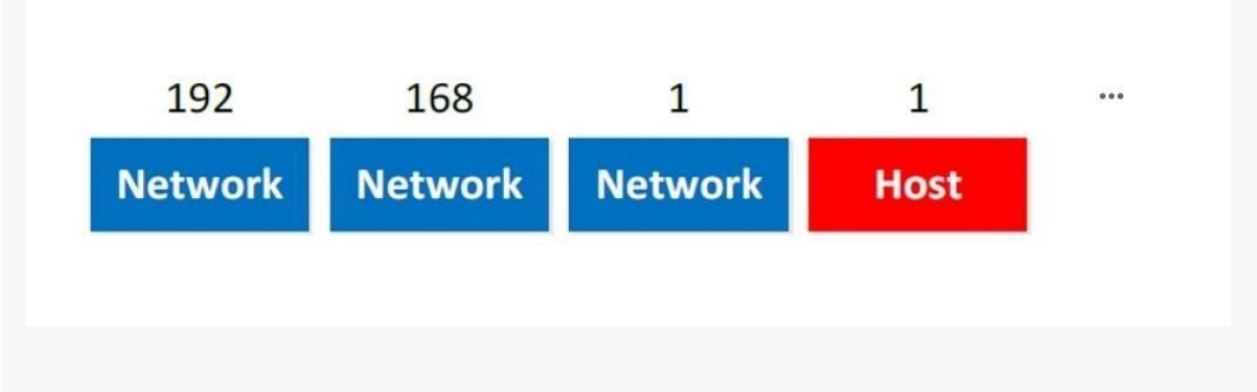
1. IP Address
2. Cisco IOS
3. Straight Cable & Cross Cable, RJ45
4. Layer2 Switch
5. Routers

S/W Requirement: - NA

**a) IP Address**.

A core function of IP address is to provide logical addressing for hosts. An IP address both uniquely identify a **host** and **netw ork** of that host. An IP address is a 32-bit number that uniquely identifies a host (computer or other device, such as a printer or router) on a TCP/IP network. IP addresses are normally expressed in dotteddecimal format, with four numbers separated by periods, such as 192.168.123.132. To understand how subnet masks are used to distinguish between hosts, networks, and subnetworks, examine an IP address in binary notation. For example, the dotted-decimal IP address 192.168.123.132 is (in binary notation) the 32 bit number 11000000010100011110111000010 0. This number may be hard to make sense of, so divide it into four parts of eight binary digits.These eight bit sections are known as octets.

The second item, which is required for TCP/IP to work, is the subnet mask. The subnet mask is used by the TCP/IP protocol to determine whether a host is on the local subnet or on a remote network. In TCP/IP, the parts of the IP address that are used as the network and host addresses are not fixed, so the network and host addresses above cannot be determined unless you have more information. This information is supplied in another 32-bit number called a subnet mask.




**IP Address Classes.**

Internet addresses are allocated by the InterNIC, the organization that administers the Internet. These IP addresses are divided into classes. The most common of these are classes A, B, and C. Classes D and E exist, but are not generally used by end users. Each of the address classes has a different default subnet mask. You can identify the class of an IP address by looking at its first octet. Following are the ranges of Class A, B, and C Internet addresses, each with an example address.

- Class A networks use a default subnet mask of 255.0.0.0 and have 0-127 as their first octet. The address 10.52.36.11 is a class A address. Its first octet is 10, which is between 1 and 126, inclusive.
- Class B networks use a default subnet mask of 255.255.0.0 and have 128-191 as their first octet. The address 172.16.52.63 is a class B address. Its first octet is 172, which is between 128 and 191, inclusive.
- Class C networks use a default subnet mask of 255.255.255.0 and have 192-223 as their first octet. The address 192.168.123.132 is a class C address. Its first octet is 192, which is between 192 and 223, inclusive.

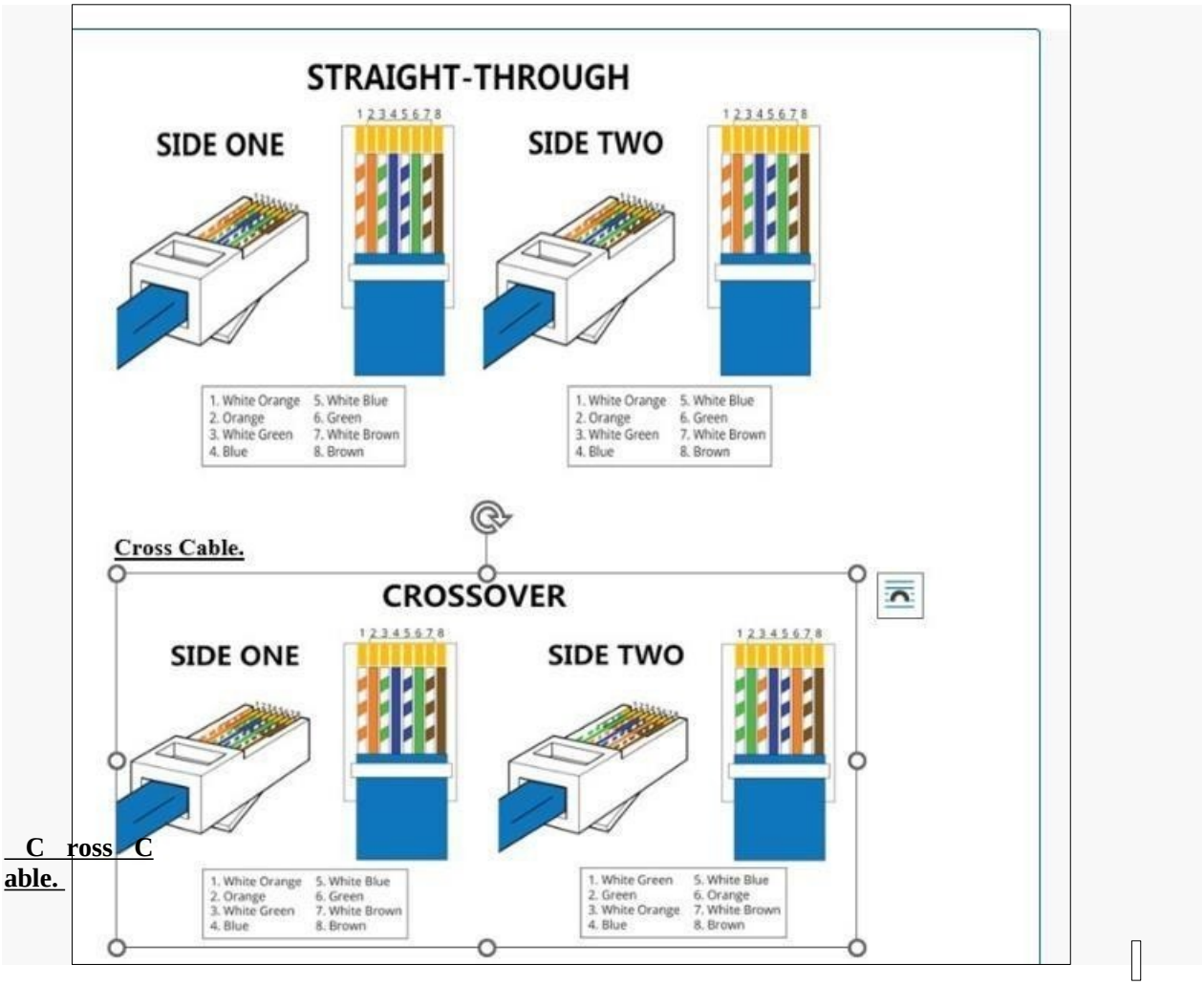
**b) Cisco IOS.**

The Cisco IOS is a command-line interface used by nearly all current Cisco Routers & Switches. Cisco IOS, formally the Internetwork Operating System, is a family of network operating systems used on many Cisco Systems routers and current Cisco network switches. Earlier, Cisco switches ran CatOS. IOS is a package of routing, switching, internetworking and telecommunications functions integrated into a multitasking operating system. Although the IOS code base includes a cooperative multitasking kernel, most IOS features have been ported to other kernels such as QNX and Linux for use in Cisco products.

**c) Straight Cable & Cross Cable, RJ45.**

The cable can be categorized as Cat 5, Cat 5e, and Cat 6 UTP cable. Cat 5 UTP cable can support 10/100 Mbps Ethernet network, whereas Cat 5e and Cat 6 UTP cable can support Ethernet network running at 10/100/1000 Mbps. You might hear about Cat 3 UTP cable, it's not popular anymore since it can only support 10 Mbps
Ethernet network. Straight and crossover cable can be Cat3, Cat 5, Cat 5e or Cat 6 UTP cable, the only difference is each type will have different wire arrangement in the cable for serving different purposes.

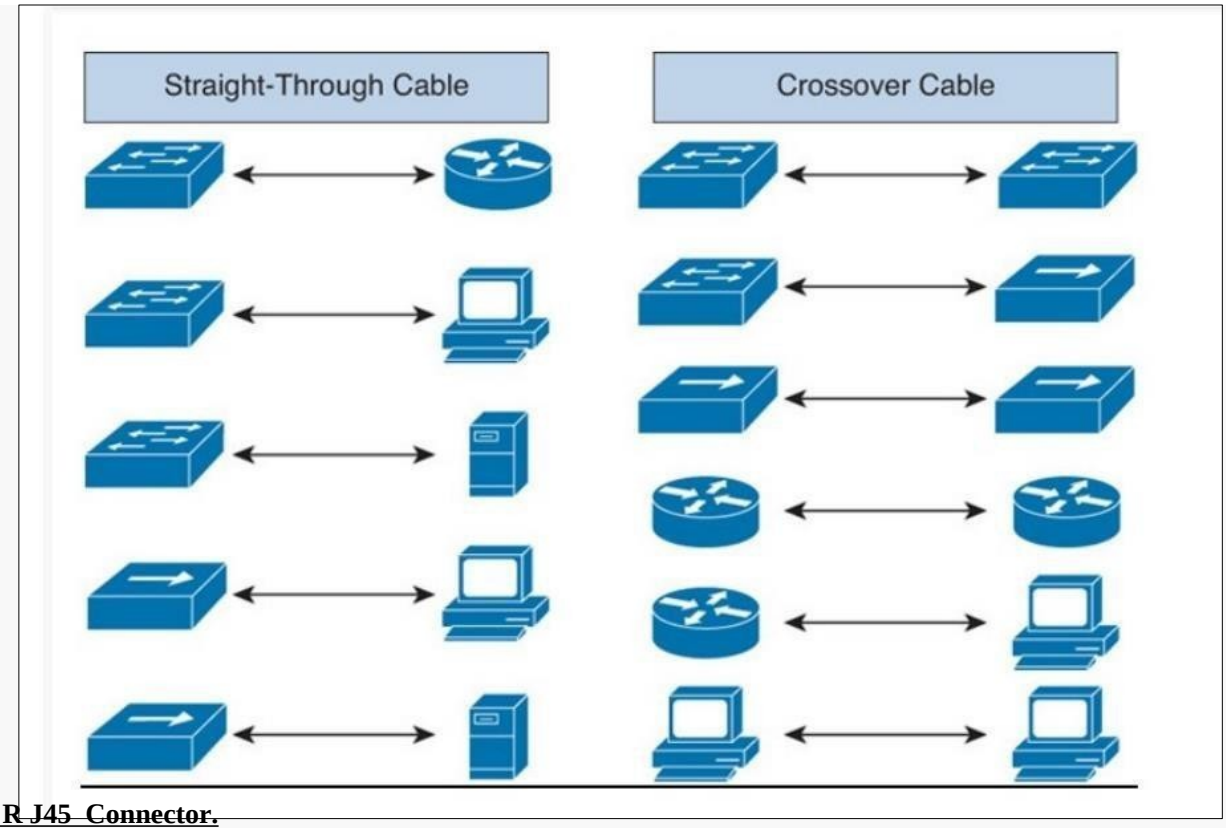
**Straight Cable.**




**Cross Cable.**




**Cross Cable.**




**R J45 Connector.**

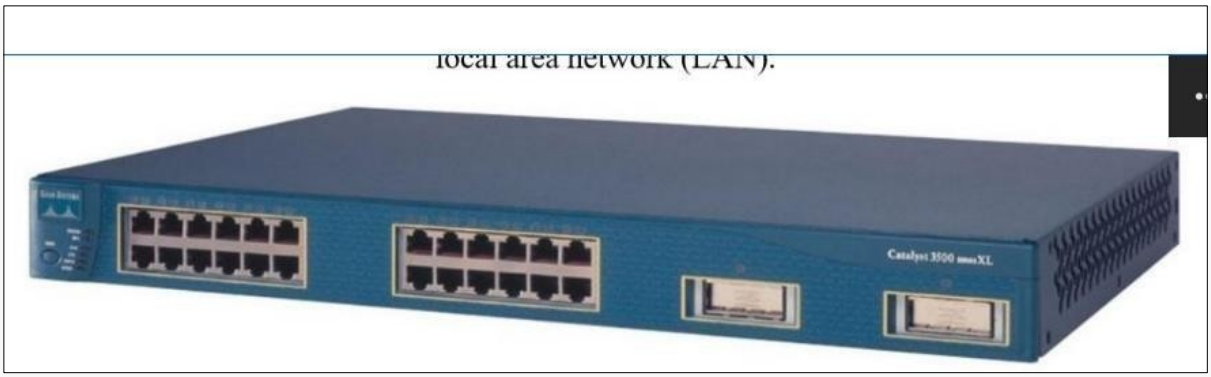
Registered Jack 45 (RJ45) is a standard type of physical connector for network cables. RJ45 connectors are most commonly seen with Ethernet cables and networks. Modern Ethernet cables feature small plastic plugs on each end that are inserted into the RJ45 jacks of Ethernet devices.



d) Layer-2 Switch.

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A layer 2 switch is a type of network switch or device that works on the data link layer (OSI Layer 2) and utilizes MAC Address to determine the path through where the frames are to be forwarded. It uses hardware based switching techniques to connect and transmit data in a local area network (LAN).



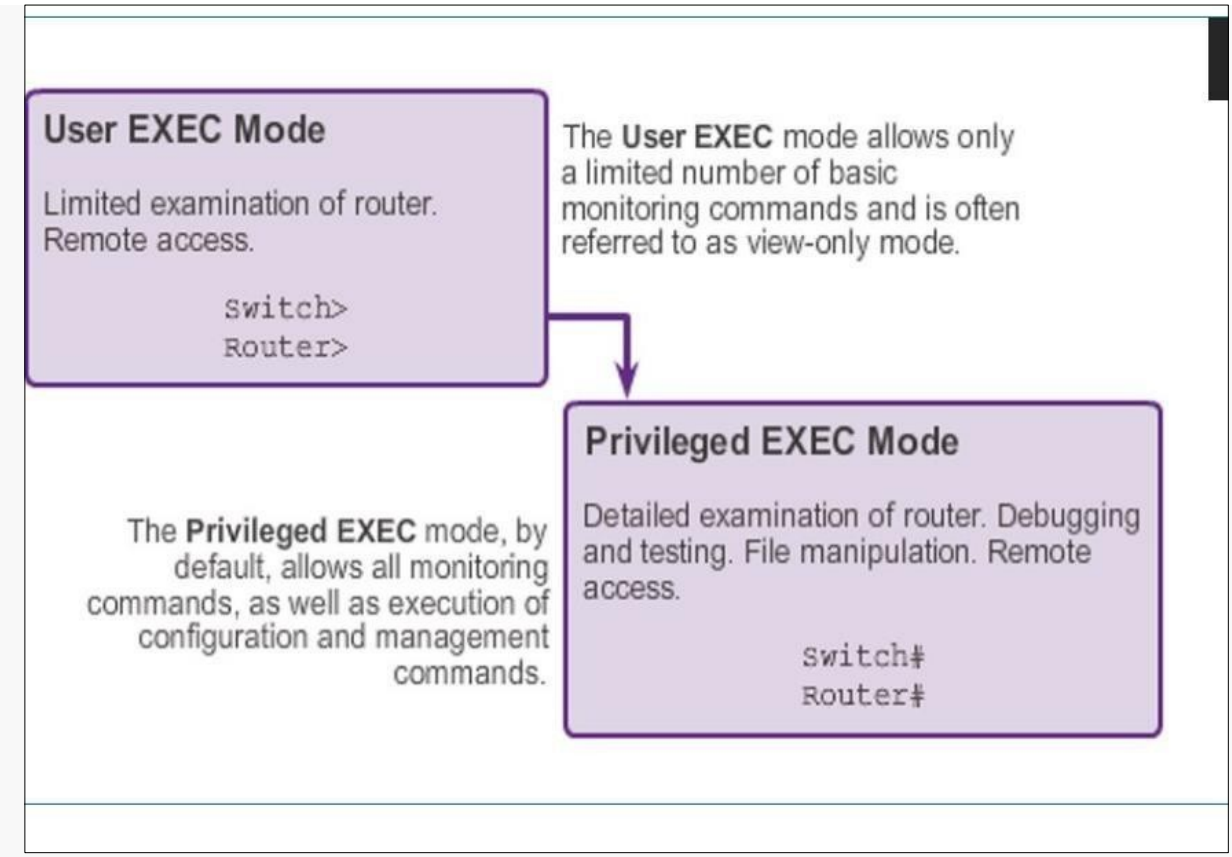

**d) Routers.**

A router is a networking device that forwards data packets between computer networks. Routers perform the traffic directing functions on the Internet. Data sent through the internet, such as a web page or email, is in the form of data packets. A packet is typically forwarded from one router to another router through the networks that constitute an internetwork (e.g. the Internet) until it reaches its destination node.

**Router Memory Components.**

| Memory | Volatile / Non-Volatile | Stores |
|---|---|---|
| **RAM** | Volatile | • Running IOS<br>• Running configuration file<br>• IP routing and ARP tables<br>• Packet buffer |
| **ROM** | Non-Volatile | • Bootup instructions<br>• Basic diagnostic software<br>• Limited IOS |
| **NVRAM** | Non-Volatile | • Startup configuration file |
| **Flash** | Non-Volatile | • IOS<br>• Other system files |

**Router Modes.**




**Further Reading**

1. https://geekflare.com/understanding-ip-address/
2. https://docs.oracle.com/cd/E19504-01/802-5753/planning3-18471/index.html
3. https://www.ibm.com/cloud/learn/networking-a-complete-guide

**Research Articles:**


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